

Search Engine Implementation Report

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March 2026

1 Introduction

This report presents the design and implementation of a scalable search engine that ranks documents using the BM25 relevance model. The system processes a collection of Wikipedia articles, building an inverted index, document lengths, and global statistics through a series of Hadoop MapReduce pipelines. The resulting index is stored in ScyllaDB for efficient retrieval, while PySpark (running on YARN) handles incremental document addition and query scoring. The implementation demonstrates a relevant application of map-reduce pipelines running on large-scale data.

2 Methodology

2.1 Technical Stack

The search engine is implemented using Apache Hadoop technologies (HDFS, Spark) for Big Data processing and storing objectives and ScyllaDB for efficient information retrieval. The Hadoop cluster and the ScyllaDB server are launched as separate Docker containers within the same network. The core functionality is performed on the master of the Hadoop cluster.

2.2 Data Loading and Preparation

This subsection outlines the key components of document preparation for indexing.

`/app/preprocess.py`

To begin with, the preprocessing pipeline functions were defined to prepare raw text for indexing using `nlTK`: `preprocess_text` and `preprocess_line`. Both functions normalize text to lowercase letters, remove punctuation, exclude English stop words, and lemmatize words using WordNet data. The difference is that `preprocess_text` returns a string of space separated words (used for preparing documents for Hadoop MapReduce pipeline) and `preprocess_line` returns a list of words (used for indexer updates using Spark RDD).

`/app/prepare_data.py`

The following [file](#) from Kaggle is loaded and used to randomly sample 1,000 documents using Spark Dataframe. The document titles are processed to have appropriate filenames compatible with the HDFS (no special characters, - handled with `sanitize_filename`, only ASCII characters, - handled with `unicode`). This stage is optional to re-run, the repository already features 1,000 documents ready to be used for processing.

`/app/preprocess_data.py`

Before invoking the scripts, the directory `/app/data` is copied to the HDFS. Next, the script reads the files using a single RDD with the `wholeTextFiles` method. The rows of a filename and a file content are mapped to the triples consisting of a document id, document title, and preprocessed with `preprocess_text` document text. Finally, the tab separated triple values are stored into `/input/data` for further MapReduce pipelines.

2.3 Indexer

This subsection summarizes key methods to create indexing statistics for BM25 scores from prepared documents using Hadoop MapReduce and store them in the database.

2.3.1 Creating Index

The Hadoop MapReduce streaming is used to prepare statistics in several stages:

Pipeline 1: This pipeline is responsible for inverted index creation.

The mapper reads the document id, document title, and document text for each row from `/input/data` and outputs a line of a word, document id, and count value 1 for each word in each document.

The sorting stage is configured to sort by the composite key of a word and a document id. The following flags were used:

```
-D stream.num.map.output.key.fields=2 \  
-D mapreduce.job.output.key.comparator.class=org.apache.hadoop.mapreduce.lib.partition.  
    KeyFieldBasedComparator \  
-D mapreduce.partition.keycomparator.options="-k1,1 -k2,2"
```

The reducer stage takes a triple of a word, a document id, and a count value, aggregates counts for each key, and outputs a key and a term frequency to the directory `/indexer/index`.

Pipeline 2: This pipeline is responsible for vocabulary creation.

The mapper reads the term, document id, and term frequency for each row from `/indexer/index` and outputs a line of a term and count value 1 for each term associated with distinct document ids.

The reducer stage takes a pair of a term and a count value, aggregates counts for each term, and outputs a key and a document frequency to the directory `/indexer/vocabulary`.

Pipeline 3: This pipeline is responsible for document length statistics.

The mapper reads the document id, document title, and document text for each row from `/input/data` and outputs a line of a document id, a document title, and count value 1 for each word in each document.

The reducer stage takes a pair of a document id and a count value, aggregates counts for each document id, and outputs a document id, its document title, and a document length to the directory `/indexer/document_lengths`.

Pipeline 4: This pipeline is responsible for global statistics (number of documents in the collection, and total length of documents in the collection).

The mapper reads the document id, document title, and document length for each row from `/indexer/document_lengths` and outputs a line of a string "key" and the document length for each document.

To ensure the output of two lines only, the following flag was employed:

```
-D mapreduce.job.reduces=1
```

The reducer stage takes a pair of the string "key" and a document length, aggregates lengths, count documents, and outputs two rows - `total_docs` with document count value and `total_doc_length` with total length of the collection - to the directory `/indexer/stats`.

2.3.2 Storing Index

To quickly extract statistics to calculate the BM25 scores, the ScyllaDB database is used as follows.
`/app/create_keyspace.py:`

The keyspace is created by connecting using `scylla-driver`. [Figure 1](#) summarizes the schema of the database.

vocabulary	
term: text	K
df: int	

inverted_index	
term: text	K
doc_id: text	C↑
tf: int	

global_stats	
key: text	K
value: double	

doc_lengths	
doc_id: text	K
doc_title: text	
dl: int	

Figure 1: ScyllaDB Tables

/app/store_index.py:

Next, all tables are loaded from the directories using PySpark and stored using the ScyllaDB connector. Note that no additional preparations are needed, since the saved formats satisfy the database schema.

2.3.3 Adding to Index

To facilitate re-indexing of new documents, the functionality was extended as follows.

add_to_index.sh

The user can specify the path to a file in the local file system, then the system generates a UUID for this document and saves the file in HDFS with the filename represented as concatenation of the generated UUID and the original filename. Following that, the next Python script is executed using PySpark.

add_to_index.py

The saved file is loaded using PySpark RDD to perform MapReduce pipeline for the term frequencies count. Before that the lines are preprocessed using `preprocess_line` function from `preprocess.py`. In addition, the document length is measured. Next, using the ScyllaDB driver, the statistics are uploaded with atomic increments where required (updates of vocabulary and global statistics): the update is performed only if the current value and the previously read value are the same (to avoid lost update). The following queries were used:

```
# Vocabulary Update
SELECT df FROM vocabulary WHERE term = ?    # read current value
UPDATE vocabulary SET df = ? WHERE term = ? IF df = ?    # update only if the value did not
change
INSERT INTO vocabulary (term, df) VALUES (?, ?) IF NOT EXISTS    # insert if term is new

# Statistics Update
SELECT value FROM global_stats WHERE key = 'total_docs'    # read current value
UPDATE global_stats SET value = ? WHERE key = 'total_docs' IF value = ?    # update only if
the value did not change

SELECT value FROM global_stats WHERE key = 'total_doc_length'
UPDATE global_stats SET value = ? WHERE key = 'total_doc_length' IF value = ?
```

2.4 Ranker

The final stage of the search engine is to rank documents with saved statistics. This stage is performed using PySpark RDD and Dataframe using Hadoop Yarn as the resource manager.

First, the system receives a query as input arguments, which are preprocessed using `preprocess_line` function. Using the PySpark ScyllaDB connector, the global statistics are retrieved and broadcasted to all nodes. Then, the vocabulary table is loaded as the dataframe and filtered for query terms.

The resulting dictionary of terms with the corresponding document frequencies is broadcasted to all nodes. Similarly, the inverted index dataframe is loaded and filtered for query terms. Finally, the document length dataframe is loaded (for further join to fetch only required rows) to complete the list of all components for the BM25 scores.

To calculate scores efficiently, the filtered inverted index and document length dataframes are transformed to RDDs and joined on the document id field. Next, each row in the resulting RDD is mapped to a triple consisting of a document id, document title, and its BM25 score (the scores are calculated according to the formula with 0.5 correction in the inverse document frequency term). Finally, the RDD is reduced by the document id using the summing of scores, and the records are sorted in the descending order. The final output is the first 10 records in the form of the document id and document title. Note that if no records were fetched, the special message is printed instead.

3 Demonstration

3.1 Installation

Ensure that Docker Engine is installed on your system. Next, clone the repository:

```
git clone https://github.com/DavidVista/okapi-bm25-pipeline
cd okapi-bm25-pipeline
```

3.2 Indexing

To start indexing of 1,000 documents, start the containers:

```
docker compose up

# Or in daemon mode
docker compose up -d
```

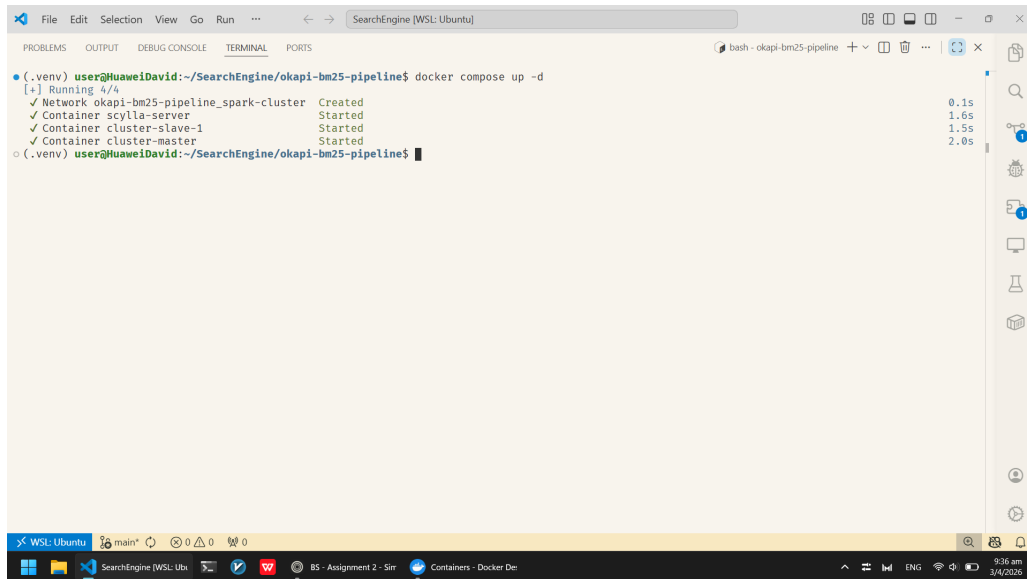


Figure 2: Starting Containers Command

3.3 Manual Launch

Below I provide commands to launch the indexing manually following each step. To make it possible, comment the following section in the `docker-compose.yml`:

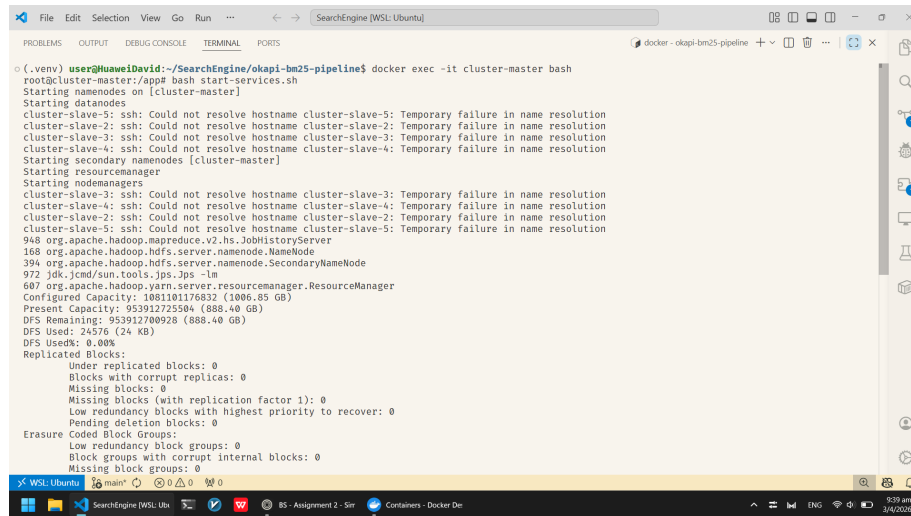
```
# You can comment the entrypoint to run the script manually inside the container
entrypoint:
```

```
- bash
- /app/app.sh
```

1. Start Hadoop Services

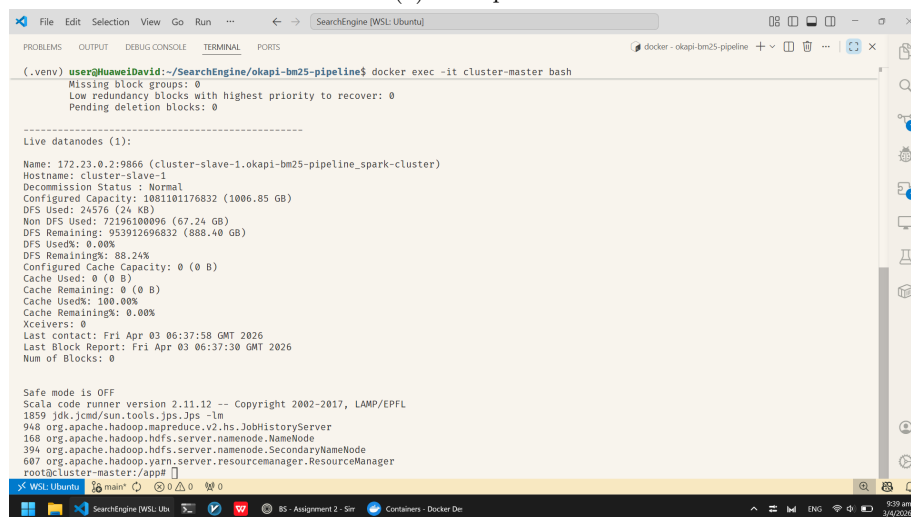
```
# Connect to cluster-master
docker exec -it cluster-master bash

# Start Hadoop services
bash start-services
```



```
(..venv) user@HuaweiDavid:~/SearchEngine/okapi-bm25-pipeline$ docker exec -it cluster-master bash
root@cluster-master:/app$ bash start-services.sh
Starting namenodes on [cluster-master]
Starting datanodes
cluster-slave-5: ssh: Could not resolve hostname cluster-slave-5: Temporary failure in name resolution
cluster-slave-2: ssh: Could not resolve hostname cluster-slave-2: Temporary failure in name resolution
cluster-slave-3: ssh: Could not resolve hostname cluster-slave-3: Temporary failure in name resolution
cluster-slave-4: ssh: Could not resolve hostname cluster-slave-4: Temporary failure in name resolution
Starting secondary namenodes [cluster-master]
Starting resource manager
Starting nodemanagers
cluster-slave-3: ssh: Could not resolve hostname cluster-slave-3: Temporary failure in name resolution
cluster-slave-4: ssh: Could not resolve hostname cluster-slave-4: Temporary failure in name resolution
cluster-slave-2: ssh: Could not resolve hostname cluster-slave-2: Temporary failure in name resolution
cluster-slave-5: ssh: Could not resolve hostname cluster-slave-5: Temporary failure in name resolution
948 org.apache.hadoop.mapreduce.v2.hs.JobHistoryServer
168 org.apache.hadoop.hdfs.server.namenode.NameNode
394 org.apache.hadoop.hdfs.server.namenode.SecondaryNameNode
972 jdk.jcmd/sun.tools.jps.Jps -lm
607 org.apache.hadoop.yarn.server.resourcemanager.ResourceManager
Configured Capacity: 108110176832 (1006.85 GB)
Present Capacity: 953912725504 (888.40 GB)
DFS Remaining: 953912700928 (888.40 GB)
DFS Used: 24576 (24 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
```

(a) First part



```
(..venv) user@HuaweiDavid:~/SearchEngine/okapi-bm25-pipeline$ docker exec -it cluster-master bash
Missing block groups: 0
Low redundancy blocks with highest priority to recover: 0
Pending deletion blocks: 0

-----
Live datanodes (1):
Name: 172.23.0.2:9866 (cluster-slave-1.okapi-bm25-pipeline_spark-cluster)
Hostname: cluster-slave-1
Decommission Status : Normal
Configured Capacity: 108110176832 (1006.85 GB)
DFS Used: 24576 (24 KB)
Non DFS Used: 72196100096 (67.24 GB)
DFS Remaining: 953912696832 (888.40 GB)
DFS Used%: 0.00%
DFS Remaining%: 88.24%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xcelfers: 0
Last contact: Fri Apr 03 06:37:58 GMT 2026
Last Block Report: Fri Apr 03 06:37:30 GMT 2026
Num of Blocks: 0

Safe mode is OFF
Scala code runner version 2.11.12 -- Copyright 2002-2017, LAMP/EPFL
1859 jdk.jcmd/sun.tools.jps.Jps -lm
948 org.apache.hadoop.mapreduce.v2.hs.JobHistoryServer
168 org.apache.hadoop.hdfs.server.namenode.NameNode
394 org.apache.hadoop.hdfs.server.namenode.SecondaryNameNode
607 org.apache.hadoop.yarn.server.resourcemanager.ResourceManager
root@cluster-master:/app#
```

(b) Second part

Figure 3: Start Services Command

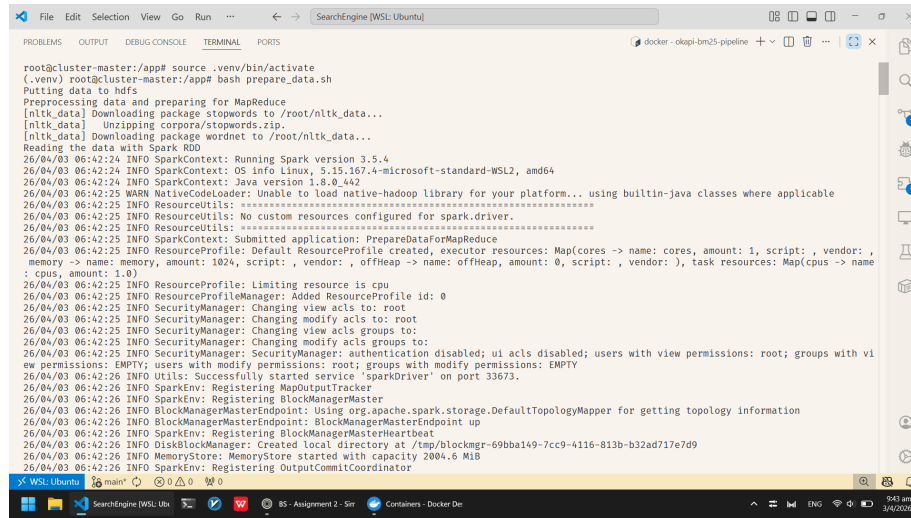
2. Start Data Preparation

```
# Creating a virtual environment
python3 -m venv .venv
source .venv/bin/activate

# Install packages
pip install -q -r requirements.txt

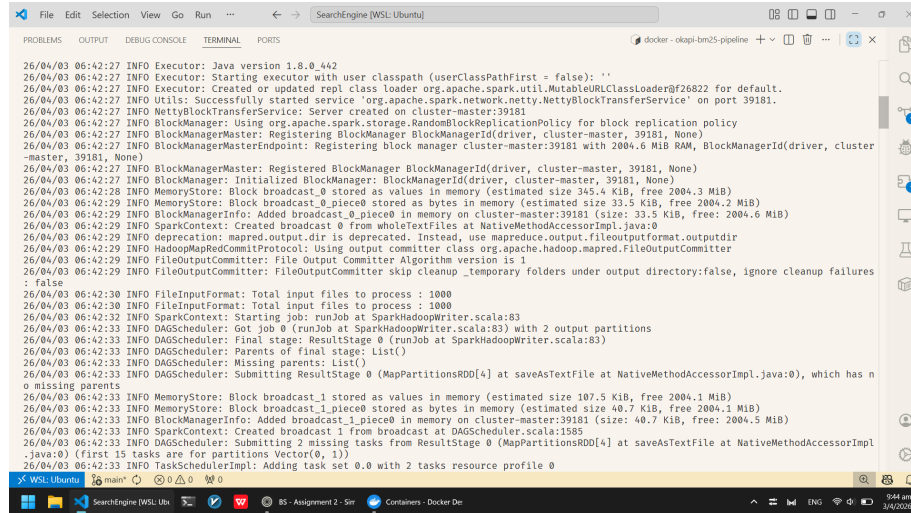
# Package the virtual env.
venv-pack -o .venv.tar.gz
```

```
# Start data preparation
bash prepare-data.sh
```



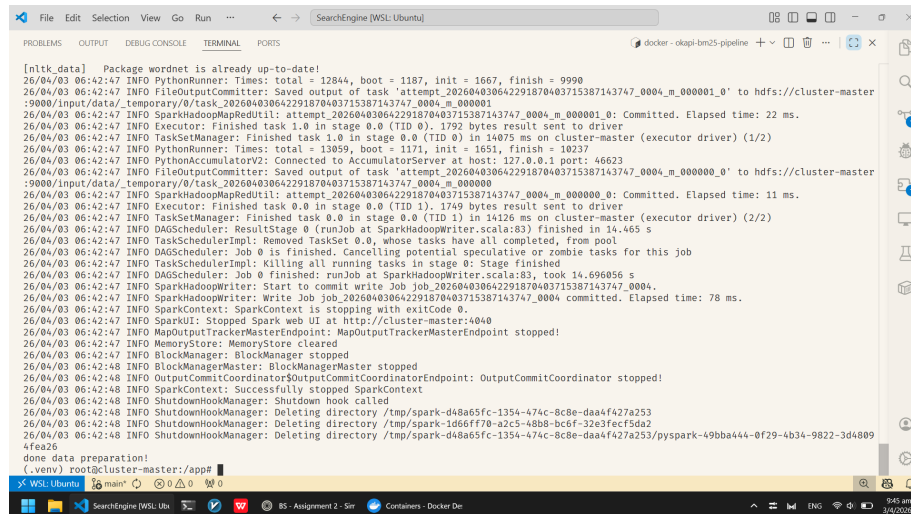
```
root@cluster-master:/app# source .venv/bin/activate
(.venv) root@cluster-master:/app# bash prepare_data.sh
Putting data to hdfs
Preprocessing data and preparing for MapReduce
[nltk_data] downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
[nltk_data] downloading package wordnet to /root/nltk_data...
Reading the data with Spark RDD
26/04/03 06:42:24 INFO SparkContext: Running Spark version 3.5.4
26/04/03 06:42:24 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/04/03 06:42:25 INFO SparkContext: Java version 1.8.0_442
26/04/03 06:42:25 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/04/03 06:42:25 INFO ResourceUtils: *****
26/04/03 06:42:25 INFO ResourceUtils: No custom resources configured for spark.driver.
26/04/03 06:42:25 INFO ResourceUtils: *****
26/04/03 06:42:25 INFO SparkContext: Submitted application: PrepareDataForMapReduce
26/04/03 06:42:25 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: ,
memory -> name: memory, amount: 1024, script: , vendor: , offheap -> name: offheap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name
: cpus, amount: 1, 0)
26/04/03 06:42:25 INFO ResourceProfile: Limiting resource is cpu
26/04/03 06:42:25 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/04/03 06:42:25 INFO SecurityManager: Changing view acls to: root
26/04/03 06:42:25 INFO SecurityManager: Changing modify acls to: root
26/04/03 06:42:25 INFO SecurityManager: Changing view acls groups to:
26/04/03 06:42:25 INFO SecurityManager: Changing modify acls groups to:
26/04/03 06:42:25 INFO SecurityManager: authentication disabled; ui acls disabled; users with view permissions: root; groups with vi
ew permissions: EMPTY; users with modify permissions: root; groups with modify permissions: EMPTY
26/04/03 06:42:26 INFO Utils: Successfully started service 'sparkDriver' on port 33673.
26/04/03 06:42:26 INFO SparkEnv: Registering MapOutputTracker
26/04/03 06:42:26 INFO SparkEnv: Registering BlockManagerMaster
26/04/03 06:42:26 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/04/03 06:42:26 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/04/03 06:42:26 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/04/03 06:42:26 INFO BlockManagerMaster: Created local directory at /tmp/blockmgr-69bba149-7cc9-4116-813b-b32ad717e7d9
26/04/03 06:42:26 INFO MemoryStore: MemoryStore started with capacity 2004.6 MiB
26/04/03 06:42:26 INFO SparkEnv: Registering OutputCommitCoordinator
```

(a) Starting Execution



```
26/04/03 06:42:27 INFO Executor: Java version 1.8.0_442
26/04/03 06:42:27 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/04/03 06:42:27 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@f26822 for default.
26/04/03 06:42:27 INFO NettyBlockTransferService: Server created on cluster-master:39181
26/04/03 06:42:27 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/04/03 06:42:27 INFO BlockManagerMaster: Registering BlockManagerId(driver, cluster-master, 39181, None)
26/04/03 06:42:27 INFO BlockManagerMasterEndpoint: Registering block manager cluster-master:39181 with 2004.6 MiB RAM, BlockManagerId(driver, cluster
-master, 39181, None)
26/04/03 06:42:27 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, cluster-master, 39181, None)
26/04/03 06:42:27 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, cluster-master, 39181, None)
26/04/03 06:42:28 INFO MemoryStore: Block broadcast_0 stored as values in memory (estimated size 345.4 KiB, free 2004.3 MiB)
26/04/03 06:42:29 INFO MemoryStore: Block broadcast_0_piece0 stored as bytes in memory (estimated size 33.5 KiB, free 2004.2 MiB)
26/04/03 06:42:29 INFO BlockManagerInfo: Added broadcast_0_piece0 in memory on cluster-master:39181 (size: 33.5 KiB, free: 2004.6 MiB)
26/04/03 06:42:29 INFO SparkContext: Created broadcast 0 from wholeTextFiles at NativeMethodAccessorImpl.java:0
26/04/03 06:42:29 INFO deprecation: mapred.output.dir is deprecated. Instead, use mapreduce.output.fileoutputformat.outputdir
26/04/03 06:42:29 INFO HadoopMapRedCommitProtocol: Using output committer class org.apache.hadoop.mapred.FileOutputCommitter
26/04/03 06:42:29 INFO FileOutputCommitter: File OutputCommitter Algorithm version is 1
26/04/03 06:42:29 INFO FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures
: false
26/04/03 06:42:30 INFO FileInputFormat: Total input files to process : 1000
26/04/03 06:42:30 INFO FileInputFormat: Total input files to process : 1000
26/04/03 06:42:32 INFO SparkContext: Starting Job: runJob at SparkHadoopWriter.scala:83
26/04/03 06:42:33 INFO DAGScheduler: Got job 0 (runJob at SparkHadoopWriter.scala:83) with 2 output partitions
26/04/03 06:42:33 INFO DAGScheduler: Final stage: ResultStage 0 (runJob at SparkHadoopWriter.scala:83)
26/04/03 06:42:33 INFO DAGScheduler: Parents of final stage: List()
26/04/03 06:42:33 INFO DAGScheduler: Missing parents: List()
26/04/03 06:42:33 INFO DAGScheduler: Submitting ResultStage 0 (MapPartitionsRDD[4] at saveAsTextFile at NativeMethodAccessorImpl.java:0), which has n
o missing parents
26/04/03 06:42:33 INFO MemoryStore: Block broadcast_1 stored as values in memory (estimated size 107.5 KiB, free 2004.1 MiB)
26/04/03 06:42:33 INFO MemoryStore: Block broadcast_1_piece0 stored as bytes in memory (estimated size 40.7 KiB, free 2004.1 MiB)
26/04/03 06:42:33 INFO BlockManagerInfo: Added broadcast_1_piece0 in memory on cluster-master:39181 (size: 40.7 KiB, free: 2004.5 MiB)
26/04/03 06:42:33 INFO SparkContext: Created broadcast 1 from broadcast at DAGScheduler.scala:1585
26/04/03 06:42:33 INFO DAGScheduler: Submitting 2 missing tasks from ResultStage 0 (MapPartitionsRDD[4] at saveAsTextFile at NativeMethodAccessorImpl
.java:0) (first 15 tasks are for partitions Vector(0, 1))
26/04/03 06:42:33 INFO TaskSchedulerImpl: Adding task set 0.0 with 2 tasks resource profile 0
```

(b) Proof of Indexing of 1,000 Documents



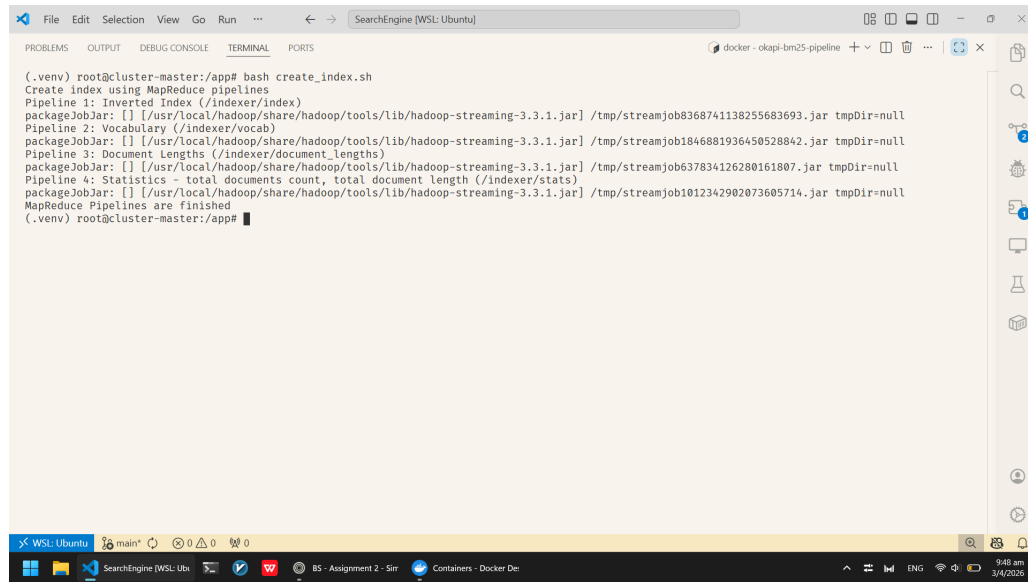
```
[nltk_data] Package wordnet is already up-to-date!
26/04/03 06:42:47 INFO PythonRunner: Times: total = 12844, boot = 1187, init = 1667, finish = 9900
26/04/03 06:42:47 INFO FileOutputCommitter: Saved output of task 'attempt_202604030642291870403715387143747_0004_m_0000001_0' to hdfs://cluster-master
:0000/input/data/_temporary/0/task_202604030642291870403715387143747_0004_m_0000001
26/04/03 06:42:47 INFO SparkHadoopMapRedUtil: attempt_202604030642291870403715387143747_0004_m_0000001_0: Committed. Elapsed time: 22 ms.
26/04/03 06:42:47 INFO Executor: Finished task 1.0 in stage 0.0 (TID 0): 1792 bytes result sent to driver
26/04/03 06:42:47 INFO TaskSetManager: Finished task 1.0 in stage 0.0 (TID 0) in 14075 ms on cluster-master (executor driver) (1/2)
26/04/03 06:42:47 INFO PythonRunner: Times: total = 13059, boot = 1171, init = 1651, finish = 10237
26/04/03 06:42:47 INFO PythonAccumulatorV2: Connected to AccumulatorServer at host: 127.0.0.1 port: 46623
26/04/03 06:42:47 INFO FileOutputCommitter: Saved output of task 'attempt_202604030642291870403715387143747_0004_m_0000000_0' to hdfs://cluster-master
:0000/input/data/_temporary/0/task_202604030642291870403715387143747_0004_m_0000000
26/04/03 06:42:47 INFO SparkHadoopMapRedUtil: attempt_202604030642291870403715387143747_0004_m_0000000_0: Committed. Elapsed time: 11 ms.
26/04/03 06:42:47 INFO Executor: Finished task 0.0 in stage 0.0 (TID 1): 1749 bytes result sent to driver
26/04/03 06:42:47 INFO TaskSetManager: Finished task 0.0 in stage 0.0 (TID 1) in 14126 ms on cluster-master (executor driver) (2/2)
26/04/03 06:42:47 INFO DAGScheduler: ResultStage 0 (runJob at SparkHadoopWriter.scala:83) finished in 14.465 s
26/04/03 06:42:47 INFO TaskSchedulerImpl: Removed TaskSet 0.0, whose tasks have all completed, from pool
26/04/03 06:42:47 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for this job
26/04/03 06:42:47 INFO TaskSchedulerImpl: Killing all running tasks in stage 0: Stage finished
26/04/03 06:42:47 INFO DAGScheduler: Job 0 finished: runJob at SparkHadoopWriter.scala:83, took 14.69606 s
26/04/03 06:42:47 INFO SparkHadoopWriter: Start to commit write Job job_202604030642291870403715387143747_0004
26/04/03 06:42:47 INFO SparkHadoopWriter: Write Job job_202604030642291870403715387143747_0004 committed. Elapsed time: 78 ms.
26/04/03 06:42:47 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/04/03 06:42:47 INFO SparkUI: Stopped spark web UI at http://cluster-master:4040
26/04/03 06:42:47 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/04/03 06:42:47 INFO MemoryStore: MemoryStore cleared
26/04/03 06:42:47 INFO BlockManager: BlockManager stopped
26/04/03 06:42:47 INFO BlockManagerMaster: BlockManagerMaster stopped
26/04/03 06:42:47 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/04/03 06:42:48 INFO SparkContext: Successfully stopped SparkContext
26/04/03 06:42:48 INFO ShutdownHookManager: Shutdown hook called
26/04/03 06:42:48 INFO ShutdownHookManager: Deleting directory /tmp/spark-d48a65fc-1354-474c-8c8e-daa4f427a253
26/04/03 06:42:48 INFO ShutdownHookManager: Deleting directory /tmp/spark-1d66ff70-a2c5-48b8-bc6f-32c3ef35da2
26/04/03 06:42:48 INFO ShutdownHookManager: Deleting directory /tmp/spark-d48a65fc-1354-474c-8c8e-daa4f427a253/pyspark-49bba444-0f29-4b34-9822-34d809
4feaz2
done data preparation!
(.venv) root@cluster-master:/app#
```

(c) Successful Completion

Figure 4: Preparing Data Command

3. Create Index (Hadoop MapReduce)

```
bash create-index.sh
```



The screenshot shows a terminal window titled "SearchEngine (WSL: Ubuntu)". The terminal output displays the execution of the script `create_index.sh` in a `(.venv) root@cluster-master:/app#` shell. The script performs the following actions:

- Prints: "Create index using MapReduce pipelines"
- Executes Pipeline 1: `Inverted Index (/indexer/index)` using `packageJobJar: [] [/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar] /tmp/streamjob8368741138255683693.jar tmpDir=null`
- Executes Pipeline 2: `Vocabulary (/indexer/vocab)` using `packageJobJar: [] [/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar] /tmp/streamjob1846881936450528842.jar tmpDir=null`
- Executes Pipeline 3: `Document Lengths (/indexer/document_lengths)` using `packageJobJar: [] [/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar] /tmp/streamjob637834126280161807.jar tmpDir=null`
- Executes Pipeline 4: `Statistics - total documents count, total document length (/indexer/stats)` using `packageJobJar: [] [/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar] /tmp/streamjob1012342902073685714.jar tmpDir=null`
- Prints: "MapReduce Pipelines are finished"

The terminal ends with the prompt `(.venv) root@cluster-master:/app#`.

Figure 5: Create Index Command

4. Store Index (ScyllaDB)

```
bash store-index.sh
```



```
(.venv) root@cluster-master:~/app# bash store_index.sh
Create keyspace
Keyspace 'indexdb' ready.
Table 'vocabulary' created (or already exists).
Table 'inverted_index' created (or already exists).
Table 'doc_lengths' created (or already exists).
Table 'global_stats' created (or already exists).
Schema setup complete.
Store the index and others to Cassandra/ScyllaDB tables
:: loading settings :: url = jar:file:/usr/local/spark/jars/ivy-2.5.1.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /root/.ivy2/cache
The jars for the packages stored in: /root/.ivy2/jars
com.scylladb#spark-scylladb-connector-2.12 added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-132165d7-bc86-481f-aeaa-0949ed6d0534;1.0
conf:: (default)
found com.scylladb#spark-scylladb-connector-2.12;4.0.0 in central
found com.scylladb#spark-scylladb-connector-driver-2.12;4.0.0 in central
found org.scala-lang.modules#scala-collection-compat-2.12;2.11.0 in central
found org.apache.cassandra#java-driver-core-shaded;4.18.1 in central
found com.datastax.oss#native-protocol;1.5.1 in central
found com.datastax.oss#java-driver-shaded-guava;25.1-jre-graal-sub-1 in central
found com.typesafe#config;1.4.1 in central
found org.slf4j#slf4j-api;1.7.26 in central
found io.dropwizard.metrics#metrics-core;4.1.18 in central
found org.hdrhistogram#HdrHistogram;2.1.12 in central
found org.reactivestreams#reactive-streams;1.0.3 in central
found org.apache.cassandra#java-driver-mapper-runtime;4.18.1 in central
found org.apache.cassandra#java-driver-query-builder;4.18.1 in central
found org.apache.commons#commons-lang3;3.10 in central
found com.thoughtworks.paranamer#paranamer;2.8 in central
found org.scala-lang#scala-reflect;2.12.19 in central
downloading https://repo1.maven.org/maven2/com/scylladb/spark-scylladb-connector-2.12/4.0.0/spark-scylladb-connector-2.12-4.0.0.jar ...
[SUCCESSFUL] com.scylladb#spark-scylladb-connector-2.12;4.0.0:spark-scylladb-connector-2.12.jar (1760ms)
downloading https://repo1.maven.org/maven2/com/scylladb/spark-scylladb-connector-driver-2.12/4.0.0/spark-scylladb-connector-driver-2.12-4.0.0.jar ...
[SUCCESSFUL] com.scylladb#spark-scylladb-connector-driver-2.12;4.0.0:spark-scylladb-connector-driver-2.12.jar (767ms)
downloading https://repo1.maven.org/maven2/org/scala-lang/modules/scala-collection-compat-2.12/2.11.0/scala-collection-compat-2.12-2.11.0.jar ...
```

(a) Starting Execution

```
26/04/03 06:50:12 INFO DataWritingSparkTask: Commit authorized for partition 0 (task 3, attempt 0, stage 3.0)
26/04/03 06:50:12 INFO DataWritingSparkTask: Committed partition 0 (task 3, attempt 0, stage 3.0)
26/04/03 06:50:12 INFO Executor: Finished task 0.0 in stage 3.0 (TID 3). 1433 bytes result sent to driver
26/04/03 06:50:12 INFO TaskSetManager: Finished task 0.0 in stage 3.0 (TID 3) in 104 ms on cluster-master (executor driver) (1/1)
26/04/03 06:50:12 INFO TaskSchedulerImpl: Removed TaskSet 3.0, whose tasks have all completed, from pool
26/04/03 06:50:12 INFO DAGScheduler: ResultStage 3 (save at NativeMethodAccessorImpl.java:8) finished in 0.135 s
26/04/03 06:50:12 INFO INFO DAGScheduler: Job 3 is finished. Cancelling potential speculative or zombie tasks for this job
26/04/03 06:50:12 INFO TaskSchedulerImpl: Killing all running tasks in stage 3: Stage finished
26/04/03 06:50:12 INFO DAGScheduler: Job 3 finished: save at NativeMethodAccessorImpl.java:8, took 0.166049 s
26/04/03 06:50:12 INFO AppendDataExec: Data source write support CassandraBulkWrite(org.apache.spark.sql.SparkSession@64b5e2ab,com.datastax.spark.connector.cql.CassandraConnector@283631e3,TableDef(indexdb,global_stats,ArrayBuffer(ColumnDef(key,PartitionKeyColumn,VarCharType)),ArrayBuffer(),Stream(ColumnDef(value,RegularColumn,DoubleType)),Stream(),false,false,Map()),WriteConf(BytesInBatch(1024),1000,Partition,LOCAL_QUORUM,false,false,5,None,TTLOption(DefaultValue),TimestampOption(DefaultValue),true,None),StructType(StructField(key,StringType,true),StructField(value,DoubleType,true)),org.apache.spark.SparkConf@6f83886) is committing.
26/04/03 06:50:12 INFO AppendDataExec: Data source write support CassandraBulkWrite(org.apache.spark.sql.SparkSession@64b5e2ab,com.datastax.spark.connector.cql.CassandraConnector@283631e3,TableDef(indexdb,global_stats,ArrayBuffer(ColumnDef(key,PartitionKeyColumn,VarCharType)),ArrayBuffer(),Stream(ColumnDef(value,RegularColumn,DoubleType)),Stream(),false,false,Map()),WriteConf(BytesInBatch(1024),1000,Partition,LOCAL_QUORUM,false,false,5,None,TTLOption(DefaultValue),TimestampOption(DefaultValue),true,None),StructType(StructField(key,StringType,true),StructField(value,DoubleType,true)),org.apache.spark.SparkConf@6f83886) committed.
26/04/03 06:50:12 INFO SparkContext: Invoking stop() from shutdown hook
26/04/03 06:50:12 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/04/03 06:50:12 INFO CassandraConnector: Disconnected from Cassandra cluster.
26/04/03 06:50:12 INFO SerialShutdownHooks: Successfully executed shutdown hook: Clearing session cache for C* connector
26/04/03 06:50:12 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
26/04/03 06:50:12 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/04/03 06:50:12 INFO MemoryStore: MemoryStore cleared
26/04/03 06:50:12 INFO BlockManager: BlockManager stopped
26/04/03 06:50:12 INFO BlockManagerMaster: BlockManagerMaster stopped
26/04/03 06:50:12 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/04/03 06:50:12 INFO SparkContext: Successfully stopped SparkContext
26/04/03 06:50:12 INFO ShutdownHookManager: Shutdown hook called
26/04/03 06:50:12 INFO ShutdownHookManager: Deleting directory /tmp/spark-488b224b-547f-42b6-b9c6-9beca9c5a1a2
26/04/03 06:50:12 INFO ShutdownHookManager: Deleting directory /tmp/spark-c69ed663-99b6-4610-ad06-eb077eabab9c/pyspark-0d7a4030-6e4c-45fb-ac6b-2d86d3c1b1d7
26/04/03 06:50:12 INFO ShutdownHookManager: Deleting directory /tmp/spark-c69ed663-99b6-4610-ad06-eb077eabab9c
```

(b) Successful Completion

```
(.venv) user@HuaweiDavid:~/SearchEngine$ docker exec -it scylla-server cqlsh
Connected to at 172.23.0.1:9042
[cqlsh 6.0.26 | Scylla 2025.4.3-0.20260205.92d2bc4b47ae | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
cqlsh> use indexdb ;
cqlsh:indexdb> select count(*) from doc_lengths ;

count
-----
1000

(1 rows)
cqlsh:indexdb> select * from inverted_index LIMIT 5;

term          | doc_id | tf
-----
dobson        | 13633480 | 1
sain          | 14404655 | 4
daredevil    | 42261662 | 1
bessus       | 12000397 | 3
ix            | 19789501 | 1

(5 rows)
cqlsh:indexdb> select * from doc_lengths LIMIT 5;

doc_id | dl | doc_title
-----
10230685 | 86 | A_Dead_Sinking_Story
27568194 | 355 | A_Hero_Ain_t_Nothing_but_a_Sandwich_film
39719446 | 235 | A_Little_Bit_of_Luck
38294693 | 202 | A_Change_Is_Gonna_Come_Jack_McDuff_album
53794980 | 201 | A_Family_Secret_Upstairs_Downstairs_
```

(c) Proof of Storing 1,000 Documents

Figure 6: Storing Data Command

5. Search with a Query

```
# Example query
bash search.sh "christianity origin"
```



```

(.venv) root@cluster-master:app# bash search.sh "christianity origin"
:: loading settings :: url = jar:file:/usr/local/spark/jars/ivy-2.5.1.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /root/.ivy2/cache
The jars for the packages stored in: /root/.ivy2/jars
com.scylladb:spark-scylladb-connector:2.12 added as a dependency
:: resolving dependencies :: org.apache.spark:spark-submit-parent-a7ba2eb1-42fc-4728-b094-8dbf997ad0b1;1.0
confs: [default]
found com.scylladb:spark-scylladb-connector:2.12:4.0.0 in central
found com.scylladb:spark-scylladb-connector-driver:2.12:4.0.0 in central
found org.scala-lang:modules:scala-collection-compat:2.12:2.11.0 in central
found org.apache.cassandra:java-driver-core-shaded:4.18.1 in central
found com.datastax.oss:native-protocol:1.5.1 in central
found com.datastax.oss:java-driver-shaded-guava:25.1-jre-graal-sub-1 in central
found com.typesafe:config:1.4.1 in central
found org.slf4j:slf4j-api:1.7.26 in central
found io.dropwizard:metrics-metrics-core:4.1.18 in central
found org.hdrhistogram:HdrHistogram:2.1.12 in central
found org.reactivestreams:reactive-streams:1.0.3 in central
found org.apache.cassandra:java-driver-mapper-runtime:4.18.1 in central
found org.apache.cassandra:java-driver-query-builder:4.18.1 in central
found org.apache.commons:commons-lang3:3.10 in central
found com.thoughtworks.paranamer:paranamer:2.8 in central
found org.scala-lang:scala-reflect:2.12.10 in central
:: resolution report :: resolve 1166ms :: artifacts dl 45ms
:: modules in use:
com.datastax.oss:java-driver-shaded-guava:25.1-jre-graal-sub-1 from central in [default]
com.datastax.oss:native-protocol:1.5.1 from central in [default]
com.scylladb:spark-scylladb-connector-driver:2.12:4.0.0 from central in [default]
com.scylladb:spark-scylladb-connector:2.12:4.0.0 from central in [default]
com.thoughtworks.paranamer:paranamer:2.8 from central in [default]
com.typesafe:config:1.4.1 from central in [default]
io.dropwizard:metrics-metrics-core:4.1.18 from central in [default]
org.apache.cassandra:java-driver-core-shaded:4.18.1 from central in [default]
org.apache.cassandra:java-driver-mapper-runtime:4.18.1 from central in [default]
org.apache.cassandra:java-driver-query-builder:4.18.1 from central in [default]
org.apache.commons:commons-lang3:3.10 from central in [default]

```

(a) First part

```

26/04/03 06:55:10 INFO ServerInfo: Adding filter to /executors: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /executors/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /executors/threadDump: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /executors/threadDump/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /static: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /api: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /jobs/job/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /stages/stage/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO ServerInfo: Adding filter to /metrics/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 06:55:10 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after waiting maxRegister
dResourcesWaitingTime: 3000000000(ms)
26/04/03 06:55:10 INFO YarnSchedulerBackend$YarnSchedulerEndpoint: ApplicationMaster registered as NettyRpcEndpointRef(spark-cl
ient://YarnAM)
Starting search for query: ['christianity', 'origin']
26/04/03 06:56:11 ERROR YarnScheduler: Lost executor 1 on cluster-slave-1: Container from a bad node: container_1775198268592_00
05_01_0000002 on host: cluster-slave-1. Exit status: 137. Diagnostics: [2026-04-03 06:56:10.845]Container killed on request. Exit
code is 137
[2026-04-03 06:56:10.845]Container exited with a non-zero exit code 137.
[2026-04-03 06:56:10.847]Killed by external signal
23481555 A_Collation_of_Four_Important_Manuscripts_of_the_Gospels
49484827 A_History_of_Christianity
59441239 A_History_of_the_Modern_World
46808416 A_Horse_Called_Bear
23141177 A_Key_into_the_Language_of_America
46692537 A_Lion_s_Trail
22844430 A_Good_Baby
2513305 A_Outset_of_Games
2847432 A_Life_on_the_Ocean_Wave
58622515 A_Boy_and_a_Priest
(.venv) root@cluster-master:app#

```

(b) Second part

Figure 7: Search a Query Command

The first three documents contain the information to elaborate on the query.

6. Add a New Document to the Index

```
# Local file A_cat.txt
bash add_to_index.sh "A_cat.txt"
```

```

(.venv) root@cluster-master:app# bash add_to_index.sh "A_cat.txt"
File A_cat.txt successfully copied to /data
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
Reading the uploaded file with Spark
26/04/03 06:57:39 INFO SparkContext: Running Spark version 3.5.4
26/04/03 06:57:39 INFO SparkContext: OS info Linux, 5.15.167.4-microsoft-standard-WSL2, amd64
26/04/03 06:57:39 INFO SparkContext: Java version 1.8.0_442
26/04/03 06:57:40 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/04/03 06:57:40 INFO ResourceUtils: =====
26/04/03 06:57:40 INFO ResourceUtils: No custom resources configured for spark.driver.
26/04/03 06:57:40 INFO SparkContext: Submitted application: WordCountRDD
26/04/03 06:57:40 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: , task resources: Map(cpu -> name: cpu, amount: 1.0)
26/04/03 06:57:40 INFO ResourceProfile: Limiting resource is cpu
26/04/03 06:57:40 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/04/03 06:57:40 INFO SecurityManager: Changing view acls to: root
26/04/03 06:57:40 INFO SecurityManager: Changing modify acls to: root
26/04/03 06:57:40 INFO SecurityManager: Changing view acls groups to:
root; groups with view permissions: root; users with view permissions:
root; groups with modify permissions: root; users with modify permissions:
EMPTY
26/04/03 06:57:41 INFO Utils: Successfully started service 'sparkDriver' on port 40905.
26/04/03 06:57:41 INFO SparkEnv: Registering MapOutputTracker
26/04/03 06:57:41 INFO SparkEnv: Registering BlockManagerMaster
26/04/03 06:57:41 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/04/03 06:57:41 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/04/03 06:57:41 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/04/03 06:57:41 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-984aec09-964d-4c3c-a371-348c72ae4ba
26/04/03 06:57:41 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB

```

(a) Starting Execution

```

26/04/03 06:57:59 INFO HadoopRDD: Input split: hdfs://cluster-master:9000/data/f1140dee-5f7a-4e97-b18a-1cf88dcde80d_A_cat.txt:0+494
26/04/03 06:58:00 INFO PythonRunner: Times: total = 65, boot = -119, init = 184, finish = 0
26/04/03 06:58:00 INFO Executor: Finished task 1.0 in stage 2.0 (TID 5). 1409 bytes result sent to driver
26/04/03 06:58:00 INFO TaskSetManager: Finished task 1.0 in stage 2.0 (TID 5) in 138 ms on cluster-master (executor driver) (1/2)
26/04/03 06:58:05 INFO PythonRunner: Times: total = 5274, boot = -220, init = 293, finish = 5201
26/04/03 06:58:05 INFO Executor: Finished task 0.0 in stage 2.0 (TID 4). 1452 bytes result sent to driver
26/04/03 06:58:05 INFO TaskSetManager: Finished task 0.0 in stage 2.0 (TID 4) in 5350 ms on cluster-master (executor driver) (2/2)
26/04/03 06:58:05 INFO TaskSchedulerImpl: Removed TaskSet 2.0, whose tasks have all completed, from pool
26/04/03 06:58:05 INFO DAGScheduler: ResultStage 2 (count at /app/add_to_index.py:35) finished in 5.379 s
26/04/03 06:58:05 INFO DAGScheduler: Job 1 is finished. Cancelling potential speculative or zombie tasks for this job
26/04/03 06:58:05 INFO TaskSchedulerImpl: Killing all running tasks in stage 2: Stage finished
26/04/03 06:58:05 INFO DAGScheduler: Job 1 finished: count at /app/add_to_index.py:35, took 7.148663 s
26/04/03 06:58:05 INFO SparkContext: SparkContext is stopping with exitCode 0
26/04/03 06:58:05 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
26/04/03 06:58:05 INFO BlockManagerInfo: Removed broadcast_3_piece0 on cluster-master:34109 in memory (size: 5.7 KiB, free: 366.3 MiB)
26/04/03 06:58:05 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/04/03 06:58:05 INFO MemoryStore: MemoryStore cleared
26/04/03 06:58:05 INFO BlockManager: BlockManager stopped
26/04/03 06:58:05 INFO BlockManagerMaster: BlockManagerMaster stopped
26/04/03 06:58:05 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/04/03 06:58:05 INFO SparkContext: Successfully stopped SparkContext
Data was prepared
Updating ScyllaDB contents
Inserted doc_lengths for f1140dee-5f7a-4e97-b18a-1cf88dcde80d_A_cat.txt, length=100
All updates completed.
Done
26/04/03 06:58:08 INFO ShutdownHookManager: Shutdown hook called
26/04/03 06:58:08 INFO ShutdownHookManager: Deleting directory /tmp/spark-1af8b120-e2ae-443c-9adc-e3103d7c8f27
26/04/03 06:58:08 INFO ShutdownHookManager: Deleting directory /tmp/spark-ffeb2a9-abab-41a1-8dc0-06dedb02f3f8
26/04/03 06:58:08 INFO ShutdownHookManager: Deleting directory /tmp/spark-1af8b120-e2ae-443c-9adc-e3103d7c8f27/pyspark-c8934b72-11f5-496a-8ff8-30348bf3580d
(.venv) root@cluster-master:app#

```

(b) Successful Completion

```

(.venv) user@HuaweiDavid:~/SearchEngine$ docker exec -it scylla-server cqlsh
Connected to at 172.23.0.1:9042
[cqlsh 6.0.26 | Scylla 2025.4.3-0.20260205.92d2bc4b7ae | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
cqlsh> use indexdb;
cqlsh:indexdb> select count(*) from doc_lengths;

+-----+
| count |
+-----+
| 1001   |
+-----+

(1 rows)
cqlsh:indexdb> select * from doc_lengths where doc_id = 'f1140dee-5f7a-4e97-b18a-1cf88dcde80d';

+-----+-----+-----+
| doc_id | dl | doc_title |
+-----+-----+-----+
| f1140dee-5f7a-4e97-b18a-1cf88dcde80d | 100 | A_cat.txt |
+-----+-----+-----+

(1 rows)
cqlsh:indexdb>

```

(c) Proof of Storing a New Document

Figure 8: Adding a new Document

Test the appearance of the new document in the search results:

```
# Example query
bash search.sh "what diet does a domestic cat have?"
```

```

(.venv) root@cluster-master:app# bash search.sh "what diet does a domestic cat have?"
:: loading settings :: url = jar:file:/usr/local/spark/jars/ivy-2.5.1.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /root/.ivy2/cache
The jars for the packages stored in: /root/.ivy2/jars
com.scylladb:spark-scylladb-connector:2.12 added as a dependency
:: resolving dependencies :: org.apache.spark:spark-submit-parent-d61bb6fa-994c-4875-94fa-9f2177fb30c2;1.0
confs: [default]
found com.scylladb:spark-scylladb-connector:2.12:4.0.0 in central
found com.scylladb:spark-scylladb-connector-driver:2.12:4.0.0 in central
found org.scala-lang.modules:scala-collection-compat:2.12:2.11.0 in central
found org.apache.cassandra:java-driver-core-shaded:4.18.1 in central
found com.datastax.oss:native-protocol:1.5.1 in central
found com.datastax.oss:java-driver-shaded-guava:25.1-jre-graal-sub-1 in central
found com.typesafe:config:1.4.1 in central
found org.slf4j:slf4j-api:1.7.26 in central
found io.dropwizard.metrics:metrics-core:4.1.18 in central
found org.hdrhistogram:HdrHistogram:2.1.12 in central
found org.reactivestreams:reactive-streams:1.0.3 in central
found org.apache.cassandra:java-driver-mapper-runtime:4.18.1 in central
found org.apache.cassandra:java-driver-query-builder:4.18.1 in central
found org.apache.commons:commons-lang3:3.10 in central
found com.thoughtworks.paranamer:paranamer:2.8 in central
found org.scala-lang:scala-reflect:2.12.10 in central
:: resolution report :: resolve 1246ms :: artifacts dl 41ms
:: modules in use:
com.datastax.oss:java-driver-shaded-guava:25.1-jre-graal-sub-1 from central in [default]
com.datastax.oss:native-protocol:1.5.1 from central in [default]
com.scylladb:spark-scylladb-connector-driver:2.12:4.0.0 from central in [default]
com.scylladb:spark-scylladb-connector:2.12:4.0.0 from central in [default]
com.thoughtworks.paranamer:paranamer:2.8 from central in [default]
com.typesafe:config:1.4.1 from central in [default]
io.dropwizard.metrics:metrics-core:4.1.18 from central in [default]
org.apache.cassandra:java-driver-core-shaded:4.18.1 from central in [default]
org.apache.cassandra:java-driver-mapper-runtime:4.18.1 from central in [default]
org.apache.cassandra:java-driver-query-builder:4.18.1 from central in [default]
org.apache.commons:commons-lang3:3.10 from central in [default]

```

(a) First part

```

26/04/03 07:01:16 INFO ServerInfo: Adding filter to /executors: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /executors/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /executors/threadDump: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /executors/threadDump/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /static: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /api: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /jobs/job/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /stages/stage/kill: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO ServerInfo: Adding filter to /metrics/json: org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/04/03 07:01:16 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after waiting maxRegister
(dResourcesWaitingTime: 3000000000(ms)
26/04/03 07:01:17 INFO YarnSchedulerBackend$YarnSchedulerEndpoint: ApplicationMaster registered as MettyRpcEndpointRef(spark-cl
ent://YarnAM)
Starting search for query: ['diet', 'domestic', 'cat']
26/04/03 07:02:15 ERROR YarnScheduler: Lost executor 1 on cluster-slave-1: Container from a bad node: container_1775198268592_00
06_01_000002 on host: cluster-slave-1. Exit status: 137. Diagnostics: [2026-04-03 07:02:14.947]Container killed on request. Exit
code is 137
[2026-04-03 07:02:14.948]Container exited with a non-zero exit code 137.
[2026-04-03 07:02:14.949]Killed by external signal
f1140dee-5f7a-4e97-b18a-1cf88cdce08d A_cat.txt
18548314 A_Cat_Abroad
42835457 A_Cat_a_Mouse_and_a_Bell
64378948 A_Gift_from_Bob
10658097 A_Dangerous_Path
9058021 A_Gunshot_to_the_Head_of_Trepidation
41682640 A_Barnyard_Frolic
35456304 A_Good_Librarian_Like_a_Good_Shepherd
40619799 A_Cure_for_Pokeritis
2393925 A_Certain_Smile
(.venv) root@cluster-master:app#

```

(b) Second part

Figure 9: Search a Query with the Updated Index

The target document (which was added to the index) is the first in the list, which is expected.

Reflection Points:

First, text normalization before indexing significantly improves the quality of search results. Specifically, punctuation removal, lowercase application, lemmatizing, and stop words removal unify both document content and query terms.

The search query script requires some time for YARN to allocate resources for the Spark job. Although local computation would be much faster, scalability would not be possible. For this example of indexing only 1,000 documents, the comparison is not possible, but generally, the approach with using RDD for computing statistics is the robust way to handle millions of documents on a cluster.